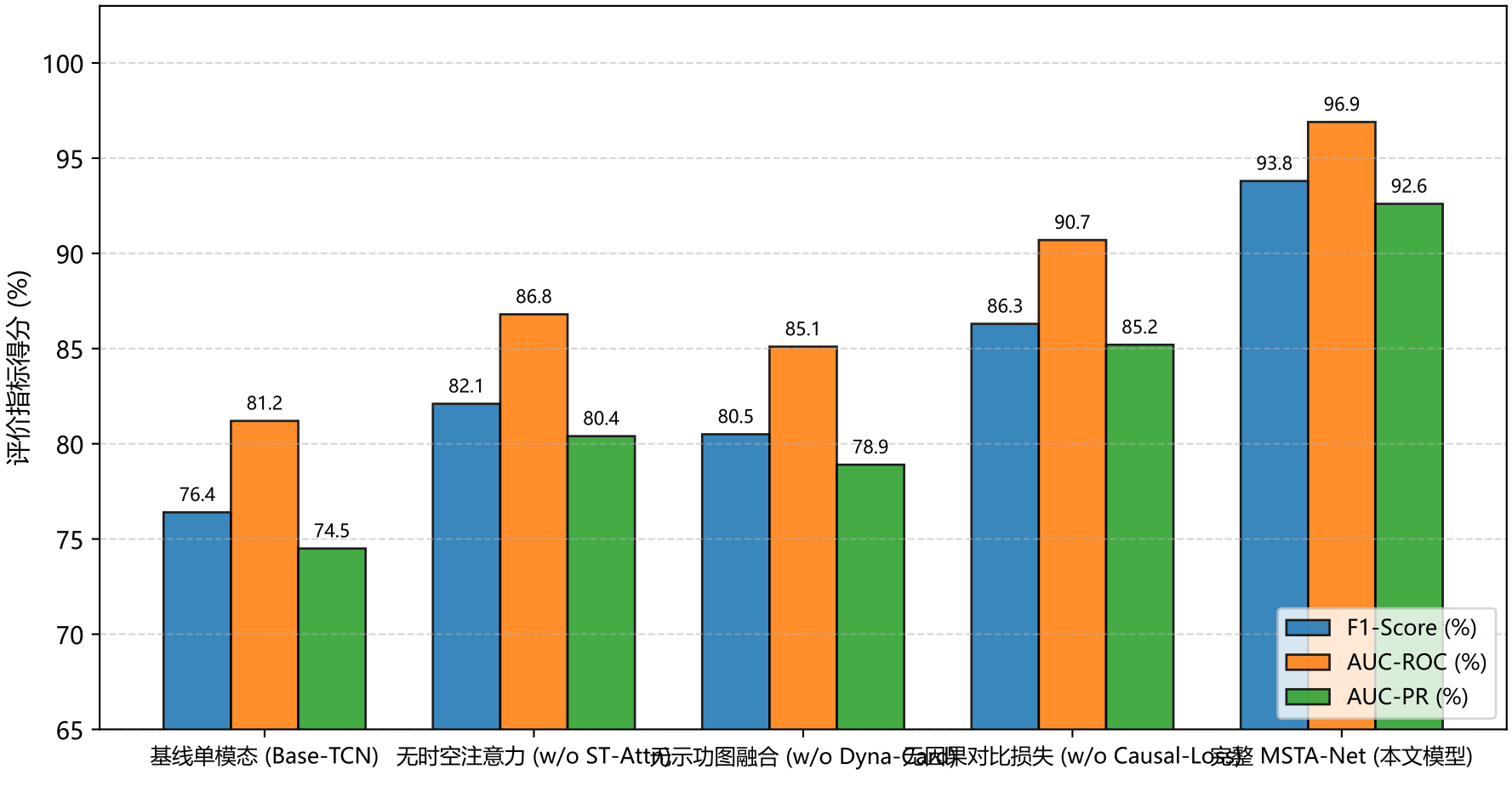
图4-2 核心模块消融实验对异常检测综合性能的影响
评价指标得分 (%)
100
95
90
85
80
75
70
65
76.4
81.2
74.5
82.1
86.8
80.4
80.5
85.1
78.9
86.3
90.7
85.2
93.8
96.9
92.6
F1-Score (%)
AUC-ROC (%)
AUC-PR (%)
基线单模态 (Base-TCN)
无时空注意力 (w/o ST-Attn)
无示功图融合 (w/o Dyna-Card)
无因果对比损失 (w/o Causal-Loss)
完整 MSTA-Net (本文模型)